Chapter 22 Biosynthesis of Amino Acids, Nucleotides, and Related Molecules

Multiple Choice Questions

1. Which of the following statements about the fixation of atmospheric nitrogen (N_2) into NH_3 by living cells is *false*?

   A) It involves the transfer of eight electrons per mol of N_2.
   B) It occurs in certain microorganisms, but not in humans.
   C) It requires a source of electrons, normally ferredoxin.
   D) It requires one ATP per mol of N_2 fixed.
   E) It requires two key protein components, each containing iron.

Overview of nitrogen metabolism
Pages: 885–886 Difficulty: 2 Ans: D

2. Which of the following is *not* true about the anammox reaction?

   A) Anammox converts ammonia to nitrogen.
   B) Anammox is performed by symbiotic bacteria of leguminous plants.
   C) Anammox generates the highly reactive molecule hydrazine that is a component of rocket fuel.
   D) Anammox occursw anaerobically.
   E) The ultimate electron acceptor in anammox is nitrite.

Overview of nitrogen metabolism
Pages: 883–884 Difficulty: 2 Ans: B

3. Which of the following enzymes is *not* involved in the assimilation of inorganic nitrogen into an organic molecule?

   A) Dinitrogenase reductase
   B) Nitrate reductase
   C) Nitrite reductase
   D) Nitrile reductase
   E) Dinitrogenase

Overview of nitrogen metabolism
Pages: 885–886 Difficulty: 2 Ans: D

4. Which of the following enzymes is *not* involved in the assimilation of inorganic nitrogen into an organic molecule?

   A) Arginase
   B) Glutamate dehydrogenase
   C) Glutamate synthase
   D) Glutamine synthetase
   E) Dinitrogenase

Overview of nitrogen metabolism
Pages: 885–889 Difficulty: 2 Ans: A

5. The enzymatic machinery to fix atmospheric N_2 into NH_4^+ is:

   A) a means of producing ATP when excess N_2 is available.
   B) composed of two key proteins, each containing iron.
   C) relatively stable when exposed to O_2.
   D) specific to plant cells.
   E) unaffected by the supply of electrons.

Overview of nitrogen metabolism
Pages: 884–886 Difficulty: 2 Ans: B

6. Glutamine synthetase converts _____ to _____ whereas glutamate synthase converts ____ to _____.

   A) formate; glutamine; ammonia; glutamate
   B) asparagine; glutamine; α-ketoglutarate; glutamate
   C) α-ketoglutarate; glutamine; oxaloacetic acid; glutamate
   D) α-ketoglutarate; glutamine; α-ketoglutarate; glutamate
   E) glutamate; glutamine; α-ketoglutarate; glutamate

Overview of nitrogen metabolism
Pages: 888–889 Difficulty: 2 Ans: E

7. Which of the following is *not* true about glutamine amidotransferases?

   A) These enzymes proceed via a covalent intermediate.
   B) Ammonia is released into a channel.
   C) Glutamate is a product of the reaction.
   D) ATP is needed to activate the glutamine.
   E) A Cys at the active site is critical for the activity of these enzymes.

Overview of nitrogen metabolism
Pages: 888–889 Difficulty: 2 Ans: D

8. Nonessential amino acids:

   A) are amino acids other than those required for protein synthesis.
   B) are not utilized in mammalian proteins.
   C) are synthesized by plants and bacteria, but not by humans.
   D) can be synthesized in humans as well as in bacteria.
   E) may be substituted with other amino acids in proteins.

Overview of nitrogen metabolism
Pages: 888–889 Difficulty: 2 Ans: D

9. An amino acid that does *not* derive its carbon skeleton, at least in part, from α-ketoglutarate is:

A) arginine.
B) glutamate.
C) glutamine.
D) proline.
E) threonine.

Biosynthesis of amino acids
Page: 892 Difficulty: 2 Ans: E

10. Glutamine, arginine, and proline:

A) do not have a common precursor.
B) may all be derived from a citric acid cycle intermediate.
C) may all be derived from a Cori cycle intermediate.
D) may all be derived from a glycolytic intermediate.
E) may all be derived from a urea cycle intermediate.

Biosynthesis of amino acids
Page: 892 Difficulty: 2 Ans: B

11. In which group are all the amino acids closely interrelated metabolically?

A) Arginine, hydroxyproline, and histidine
B) Arginine, tyrosine, and glutamate
C) Glycine, valine, glutamine, and aspartate
D) Ornithine, alanine, glycine, and valine
E) Ornithine, proline, arginine, and glutamate

Biosynthesis of amino acids
Page: 892 Difficulty: 2 Ans: E

12. If glucose labeled with ^{14}C at C-1 were the starting material for amino acid biosynthesis, the product(s) that would be readily formed is (are):

A) serine labeled at the carboxyl carbon.
B) serine labeled at alpha carbon.
C) serine labeled at the R-group carbon.
D) All of the above
E) None of the above

Biosynthesis of amino acids
Page: 894 Difficulty: 3 Ans: C

13. An amino acid that does *not* derive its carbon skeleton, at least in part, from oxaloacetate is:

A) aspartate.
B) lysine.
C) methionine.
D) proline.
E) threonine.

Biosynthesis of amino acids
Pages: 895–896 Difficulty: 3 Ans: D

14. Homoserine is:

A) a precursor of both methionine and threonine.
B) a precursor of serine.
C) derived from homocysteine.
D) derived from serine.
E) derived from threonine.

Biosynthesis of amino acids
Page: 896 Difficulty: 2 Ans: A

15. If a cell were unable to synthesize or obtain tetrahydrofolic acid (H_4 folate), it would probably be deficient in the biosynthesis of:

A) isoleucine.
B) leucine.
C) lysine.
D) methionine.
E) serine.

Biosynthesis of amino acids
Page: 896 Difficulty: 2 Ans: D

16. The nitrogen atom in the side chain of lysine is derived from which amino acid?

A) Aspartic acid
B) Glutamic acid
C) Glutamine
D) Asparagine
E) Arginine

Biosynthesis of amino acids
Page: 896 Difficulty: 2 Ans: B

17. The nitrogen atom in the indole ring of tryptophan is derived from which amino acid?

A) Aspartic acid
B) Glutamic acid
C) Glutamine
D) Asparagine
E) Arginine

Biosynthesis of amino acids
Page: 898 Difficulty: 2 Ans: C

18. Erythrose 4-phosphate is a precursor of:

A) aspartate.

B) cysteine.
C) phenylalanine.
D) serine.
E) threonine.

Biosynthesis of amino acids
Page: 898 Difficulty: 2 Ans: C

19. An important intermediate in the biosynthetic pathway to aromatic amino acids is:

A) benzoic acid.
B) lactate.
C) orotate.
D) shikimate.
E) α-ketoglutarate.

Biosynthesis of amino acids
Page: 898 Difficulty: 2 Ans: D

20. Which of the following amino acids derives its nitrogen from a purine ring?

A) Histidine
B) Lysine
C) Arginine
D) Glutamine
E) Tryptophan

Biosynthesis of amino acids
Pages: 898–899 Difficulty: 2 Ans: A

21. The amino acid _____ and is an intermediate in the biosynthesis of _____.

A) histidine; purines
B) glycine; heme
C) serine; heme
D) serine; sphingosine
E) glutamine; glutathione

Molecules derived from amino acids
Pages: 902–907 Difficulty: 2 Ans: B

22. Bile pigments are:

A) formed in the degradation of heme.
B) generated by oxidation of sterols.
C) responsible for light reception in the vertebrate eye.
D) secreted from the pancreas
E) the products of purine degradation.

Molecules derived from amino acids

Pages: 904–906 Difficulty: 1 Ans: A

23. Glutathione is a(n):

A) enzyme essential in the synthesis of glutamate.
B) isomer of oxidized glutamic acid.
C) methyl-group donor in many biosynthetic pathways.
D) product of glutamate and methionine.
E) tripeptide of glycine, glutamate, and cysteine.

Molecules derived from amino acids
Pages: 906–907 Difficulty: 2 Ans: E

24. The plant hormone indole-3-acetate (auxin) is formed from:

A) arginine.
B) histidine.
C) phenylalanine.
D) threonine.
E) tryptophan.

Molecules derived from amino acids
Page: 908 Difficulty: 2 Ans: E

25. L-Dopa is an intermediate in the conversion of:

A) phenylalanine to homogentisic acid.
B) phenylalanine to tyrosine.
C) tyrosine to epinephrine.
D) tyrosine to phenylalanine.
E) tyrosine to phenylpyruvate.

Molecules derived from amino acids
Pages: 909–910 Difficulty: 2 Ans: C

26. The amino acid that gives rise to the biological messenger NO is:

A) glutamine.
B) arginine.
C) proline.
D) lysine.
E) histidine.

Molecules derived from amino acids
Pages: 909–911 Difficulty: 2 Ans: B

27. The hormones epinephrine and norepinephrine are derived biosynthetically from:

A) arginine.
B) histidine.
C) isoleucine.

D) tryptophan.
E) tyrosine.

Molecules derived from amino acids
Pages: 909–910 Difficulty: 2 Ans: E

28. One amino acid directly involved in the purine biosynthetic pathway is:

A) alanine.
B) aspartate.
C) glutamate.
D) leucine.
E) tryptophan

Biosynthesis and degradation of nucleotides
Page: 913 Difficulty: 2 Ans: B

29. 5-Phosphoribosyl-α-pyrophosphate (PRPP) is a synthetic precursor for all of the following *except*:

A) AMP.
B) arginine.
C) histidine.
D) tryptophan.
E) UMP.

Biosynthesis and degradation of nucleotides
Page: 913 Difficulty: 3 Ans: B

30. Glutamine is a nitrogen donor in the synthesis of:

A) CTP.
B) dTTP.
C) inosinic acid (IMP).
D) orotate.
E) UMP.

Biosynthesis and degradation of nucleotides
Pages: 913–915 Difficulty: 2 Ans: C

31. De novo purine biosynthesis is distinguished from de novo pyrimidine biosynthesis by:

A) condensation of the completed purine ring with ribose phosphate
B) incorporation of CO_2.
C) inhibition by azaserine (a glutamine analog).
D) participation of aspartate.
E) participation of PRPP (phosphoribosyl pyrophosphate).

Biosynthesis and degradation of nucleotides
Pages: 913–915 Difficulty: 3 Ans: B

32. The ribosyl phosphate moiety needed for the synthesis of orotidylate, inosinate, and guanylate is

provided most directly by:

A) 5-phosphoribosyl 1-pyrophosphate.
B) adenosine 5′-phosphate.
C) guanosine 5′-phosphate.
D) ribose 5-phosphate.
E) ribulose 5-phosphate.

Biosynthesis and degradation of nucleotides
Pages: 913–915Difficulty: 2 Ans: A

33. The synthesis of purine and pyrimidine nucleotides differ in that:

A) ATP is required in the synthesis of purines but not in the synthesis of pyrimidines.
B) purine biosynthesis starts with the formation of PRPP, whereas pyrimidines incorporate the PRPP near the end of the pathway.
C) purine formation requires a THF derivative, whereas pyrimidine formation does not.
D) pyrimidine biosynthesis is tightly regulated in the cell, whereas purine biosynthesis is not.
E) pyrimidines go through many steps, adding a single carbon or nitrogen each time, whereas the basic skeleton for purines is formed by two main precursors.

Biosynthesis and degradation of nucleotides
Pages: 913–915 Difficulty: 3 Ans: B

34. Which one of the following statements is *true* of the biosynthetic pathway for purine nucleotides?

A) CO_2 does not participate in any of the steps in this pathway.
B) Deoxyribonucleotides are formed from 5-phosphodeoxyribosyl 1-pyrophosphate.
C) Inosinate is the purine nucleotide that is the precursor of both adenylate and guanylate.
D) Orotic acid is an essential precursor for purine nucleotides.
E) The amino acid valine is one of the precursors contributing to purine nucleotides.

Biosynthesis and degradation of nucleotides
Page: 913 Difficulty: 2 Ans: C

35. Orotic aciduria is an inherited metabolic disease in which orotic acid (orotate) accumulates in the tissues, blood, and urine. The metabolic pathway in which the enzyme defect occurs is:

A) epinephrine synthesis.
B) purine breakdown.
C) purine synthesis.
D) pyrimidine breakdown.
E) pyrimidine synthesis.

Biosynthesis and degradation of nucleotides
Page: 915 Difficulty: 2 Ans: E

36. Precursors for the biosynthesis of the pyrimidine ring system include:

A) carbamoyl phosphate and aspartate.
B) glutamate, NH_3, and CO_2.

C) glycine and succinyl-CoA.
D) glycine, glutamine, CO_2, and aspartate.
E) inosine and aspartate.

Biosynthesis and degradation of nucleotides
Page: 915 Difficulty: 2 Ans: A

37. The most direct precursors of the nitrogens of UMP are:

A) aspartate and carbamoyl phosphate.
B) glutamate and aspartate.
C) glutamate and carbamoyl phosphate.
D) glutamine and aspartate.
E) glutamine and carbamoyl phosphate.

Biosynthesis and degradation of nucleotides
Page: 915 Difficulty: 2 Ans: A

38. CMP, UMP, and TMP all have _____ as a common precursor.

A) adenosine
B) aspartate
C) glutamine
D) inosine
E) *S*-adenosyl methionine

Biosynthesis and degradation of nucleotides
Page: 915 Difficulty: 2 Ans: B

39. Which of the following is *not* true of the reaction catalyzed by ribonucleotide reductase?

A) Glutathione is part of the path of electron transfer.
B) It acts on nucleoside diphosphates.
C) Its mechanism involves formation of a free radical.
D) There is a separate enzyme for each nucleotide (ADP, CDP, GDP, UDP).
E) Thioredoxin acts as an essential electron carrier.

Biosynthesis and degradation of nucleotides
Pages: 917–919 Difficulty: 2 Ans: D

40. Which one of the following statements *correctly* describes the biosynthetic pathway for purine nucleotides?

A) Purine deoxynucleotides are made by the same path as ribonucleotides, followed by reduction of the ribose moiety.
B) The first enzyme in the path is aspartate transcarbamoylase (ATCase).
C) The nitrogen in the purine base that is bonded to ribose in the nucleotide is derived originally from glycine.
D) The pathway occurs only in plants and bacteria, not in animals.
E) The purine rings are first synthesized, then condensed with ribose phosphate.

Biosynthesis and degradation of nucleotides
Page: 913 Difficulty: 2 Ans: A

41. A cell that is unable to synthesize or obtain tetrahydrofolic acid (H_4 folate) would probably be deficient in the biosynthesis of:

A) CMP.
B) GMP.
C) orotate.
D) thymidylate (TMP).
E) UMP.

Biosynthesis and degradation of nucleotides
Page: 920 Difficulty 3 Ans: D

42. An intermediate of purine degradation in humans is:

A) glutamate.
B) NH_4^+.
C) succinate.
D) urea.
E) uric acid.

Biosynthesis and degradation of nucleotides
Page: 921 Difficulty: 2 Ans: E

43. Which of the following statements is *true* about gout?

A) Gout is caused by excessive breakdown of pyrimidine nucleotides.
B) Gout can be treated by administration of fluorouracil.
C) Gout is particularly prevalent in patients lacking xanthine oxidase.
D) Gout leads to deposition of sodium urate crystals in the joints.
E) Eating more liver is an effective way to treat the symptoms of gout.

Biosynthesis and degradation of nucleotides
Pages: 922–923Difficulty: 2 Ans: D

44. Which of the following is *not* a chemotherapeutic drug?

A) Azaserine
B) Allopurinol
C) Fluorouracil
D) Methotrexate
E) Acivicin

Biosynthesis and degradation of nucleotides
Page: 921 Difficulty: 2 Ans: B

Short Answer Questions

45. Trace the path of nitrogen from atmospheric N_2 into glutamate. Name the intermediates (no structures

necessary) and enzymes, and show any coenzymes involved.

Ans: First, molecular nitrogen (N_2) is reduced to ammonia in the reaction catalyzed by the nitrogenase complex, which is present in certain prokaryotes, including some that live symbiotically with legumes:

$$N_2 + 10H^+ + 8e^- + 16ATP \longrightarrow 2NH_4^+ + 16ADP + 16P_i + H_2$$

Then ammonia is incorporated into glutamine in the reaction catalyzed by glutamine synthetase:

$$\text{Glutamate} + NH_4^+ + ATP \longrightarrow \text{glutamine} + ADP + P_i + H^+$$

Finally, glutamate synthase catalyzes formation of glutamate from glutamine:

$$\alpha\text{-Ketoglutarate} + \text{glutamine} + NADPH + H^+ \longrightarrow 2\ \text{glutamate} + NADP^+$$

An alternative minor route from ammonia to glutamate involves the reaction catalyzed by glutamate dehydrogenase:

$$\alpha\text{-Ketoglutarate} + NH_4^+ + NADPH \longrightarrow \text{glutamate} + NADP^+ + H_2O$$

Overview of nitrogen metabolism
Pages: 882–888 Difficulty: 3

46. Give the overall reaction that results from the combined action of glutamate synthase and glutamine synthetase.

Ans: α-Ketoglutarate + NH_4^+ + NADPH + ATP $\longrightarrow$ glutamate + $NADP^+$ + ADP + P_i

Overview of nitrogen metabolism
Page: 888 Difficulty: 2

47. Give the equations for the two-step reaction sequence catalyzed by glutamine synthetase.

Ans:
(1) Glutamate + ATP $\longrightarrow$ γ–glutamyl phosphate + ADP
(2) γ-Glutamyl phosphate + NH_4^+ $\longrightarrow$ glutamine + P_i + H^+

Overview of nitrogen metabolism
Page: 888 Difficulty: 2

48. Describe two types of regulation of the enzyme glutamine synthetase and explain why the regulation of this enzyme is so complex.

Ans: (1) The enzyme is subject to cumulative allosteric inhibition by six compounds: AMP, tryptophan, carbamoyl phosphate, CTP, histidine, and glucosamine 6-phosphate, all of which are end products of pathways in which glutamine is a key precursor. Alanine and glycine are also allosteric inhibitors of the enzyme. (2) The enzyme is also subject to regulation by covalent alteration: adenylylation and deadenylylation. Adenylylation, which inhibits the enzyme, is indirectly stimulated by glutamine and P_i and inhibited by α-ketoglutarate and ATP. The complexity of the regulation reflects the fact that glutamine is involved in many synthetic pathways, and its level must be responsive to the

concentrations of end-products and precursors of each pathway.

Overview of nitrogen metabolism
Pages: 889–890 Difficulty: 3

49. Why is it necessary to have protein in our (human) diets?

Ans: Protein provides the 10 essential amino acids that humans cannot synthesize. We need these amino acids for protein synthesis and for the production of a variety of products, such as histamine and serotonin, derived from the essential amino acids.

Overview of nitrogen metabolism
Page**: 891 Difficulty: 2**

50. Give the name and structure of the glycolytic or citric acid cycle intermediate that has the same carbon skeleton as (a) alanine, (b) glutamate, (c) aspartate.

Ans:

(a) Pyruvate $CH_3—CO—COO^-$
(b) α-Ketoglutarate $^-OOC—CH_2—CH_2—CO—COO^-$
(c) Oxaloacetate $^-OOC—CH_2—CO—COO^-$

Biosynthesis of amino acids
Pages: 891–896 Difficulty: 2

51. Show the biosynthetic pathway for the conversion of a citric acid cycle intermediate into proline. Indicate where any cofactors participate.

Ans: α-Ketoglutarate $\rightarrow$ glutamate $\rightarrow$ γ–glutamyl phosphate $\rightarrow$ glutamate γ-semialdehyde $\rightarrow$ pyrroline-5-carboxylate $\rightarrow$ proline

For enzymes and cofactors, see Fig. 22-12.

Biosynthesis of amino acids
Page: 893 Difficulty: 3

52. Show the reaction catalyzed by glycine synthase, indicating the role of any cofactors that participate.

Ans: $CO_2 + NH_4^+ + NADH + H^+ + N^5,N^{10}$-methylenetetrahydrofolate $\longrightarrow$ glycine + NAD^+ + tetrahydrofolate

Biosynthesis of amino acids
Page: 894 Difficulty: 2

53. An animal cell is capable of converting alanine into serine. What is the shortest pathway using known enzymes by which this conversion could be accomplished? Show intermediates and cofactors; no enzyme names are required. (*Hint:* The first step is removal of the nitrogen by transamination.)

Ans: (1) Alanine $\longrightarrow$ pyruvate
(2) Pyruvate + HCO_3^- + ATP $\longrightarrow$ oxaloacetate + ADP + P_i

(3) Oxaloacetate + GTP ⟶ phosphoenolpyruvate + CO_2 + GDP
(4) Phosphoenolpyruvate ⟶ 2-phosphoglycerate ⟶ 3-phosphoglycerate;
3-phosphoglycerate + NAD^+ ⟶ 3-phosphohydroxypyruvate + NADH
(5) 3-Phosphohydroxypyruvate + glutamate ⟶ 3-phosphoserine + α-ketoglutarate
(6) 3-Phosphoserine ⟶ serine + P_i

(See Fig. 22-14, and Fig. 14-16.)

Biosynthesis of amino acids
Page: 894 Difficulty: 3

54. Show the steps by which an intermediate of glycolysis can be converted into serine.

Ans: Serine is derived from 3-phosphoglycerate by the pathway shown in Fig. 22-14. Phosphoglycerate is oxidized by transamination from glutamate, yielding 3-phosphoserine. Removal of the phosphate yields serine.

Biosynthesis of amino acids
Page: 894 Difficulty: 3

55. Show the two-step reaction catalyzed by tryptophan synthetase.

Ans:
(1) Indole-3-glycerol phosphate ⟶ indole + glyceraldehyde 3-phosphate
(2) Indole + serine ⟶ tryptophan + H_2O

(See details in Fig. 22-20.)

Biosynthesis of amino acids
Page: 901 Difficulty: 3

56. In bacteria, the amino acids listed below can be derived directly or indirectly from serine, alanine, aspartate, glutamate, or chorismate. Indicate below which of these "parent" compounds provides the carbon skeleton for each amino acid:

	Parent compound
Asparagine	_____
Tryptophan	_____
Glycine	_____
Methionine	_____
Threonine	_____
Cysteine	_____
Proline	_____
Isoleucine	_____
Phenylalanine	_____

Ans: Aspartate (asparagine); chorismate (tryptophan); serine (glycine); aspartate (methionine); aspartate (threonine); serine (cysteine); glutamate (proline); alanine via pyruate (isoleucine); chorismate (phenylalanine)

Biosynthesis of amino acids
Pages: 892–898 Difficulty: 3

57. Describe and contrast, with diagrams, concerted (cumulative) feedback regulation and sequential feedback inhibition.

Ans: Both are allosteric mechanisms to reduce the flux through a pathway when the end product(s) are present in sufficient quantities. In concerted feedback regulation, each of several end products partially inhibits the first enzyme in the pathway. Glutamine synthetase exhibits this kind of regulation (Fig. 22-8). Sequential feedback inhibition occurs in a branched pathway; the product of each branch inhibits the first step in that branch. When both end products are present, the last intermediate before the branch accumulates, and it feeds back on (inhibits) the first enzyme in the main path. The synthesis of AMP and GMP (Fig. 22-37) exemplifies this mechanism. Sequential feedback inhibition is also seen in pathways where there is enzyme multiplicity, thus preventing one biosynthetic end product from shutting down key steps in the pathway when other end products might still be needed. The synthesis and control of lysine, methionine, threonine, and isoleucine exemplifies this mechanism (Fig. 22-24).

Biosynthesis of amino acids
Pages: 899–900, 904, 914 Difficulty: 3

58. Show how you can form δ-aminolevulinate from either glycine and succinyl-CoA or from glutamate.

Ans: In most animals, glycine and succinyl-CoA condense to form α-amino-β-ketoadipate that is decarboxylated to form δ-aminolevulinate (Fig. 22-25a). In bacteria and plants, glutamate forms a glutamyl-tRNA that is subsequently reduced and cleaved from the tRNA. Finally, an amidotransferase moves the nitrogen to form form δ-aminolevulinate. (See Fig. 22-25b.)

Molecules derived from amino acids
Page: 905 Difficulty: 3

59. Match the signaling molecule with its amino acid precursor (a given precursor may be used more than once or not at all):

Signal Molecule	Amino Acid Precursor
(a) auxin	(1) histidine
(b) epinephrine	(2) glutamic acid
(c) γ-amino butyrate	(3) tyrosine
(d) histamine	(4) tryptophan
(e) serotonin	(5) arginine
(f) NO	

Ans: (a)-(4); (b)-(3); (c)-(2); (d)-(1); (e)-(4); (f)-(5)

Molecules derived from amino acids
Pages: 908–909 Difficulty: 2

60. Draw the structure of 5′-IMP. Indicate with arrows those carbon atoms donated by derivatives of tetrahydrofolate and circle the atoms derived from glycine.

Ans: The structure of 5′-IMP with the origin of each atom is shown in Figs. 22-34 and 22-35.

Biosynthesis and degradation of nucleotides
Page: 912 Difficulty: 3

61. Draw the structure of 5′-GMP. Indicate with an arrow the atom(s) derived from glutamine's amide group(s).

    Ans: The structure of 5′-GMP is shown in Fig. 22-36. The atoms derived from glutamine's amide group are the two "bottom" ring nitrogens (see Fig. 22-25) and the external ring nitrogen (see Fig. 22-36).

Biosynthesis and degradation of nucleotides
Pages: 913–914 Difficulty: 3

62. Draw the structure of 5′AMP. What is the first "committed" step in the biosynthetic sequence that leads to 5′AMP? How is this step regulated?

    Ans: The structure of AMP is in Fig. 22-36. The first committed step in the pathway to AMP is the formation of 5-phosphoribosylamine from PRPP. (See Fig. 22-35.) This step is regulated by feedback inhibition; accumulated AMP, IMP, and GMP allosterically inhibit the first enzyme in the path, as shown in Fig. 22-37.

Biosynthesis and degradation of nucleotides
Pages: 913–914 Difficulty: 3

63. Draw the structure of 5′-UMP (uridylic acid). Circle those carbon atoms donated by atoms derived from aspartate.

    Ans: The structure of UMP is shown in Fig. 22-38, which also shows the atoms contributed by aspartate in the first step of the pathway to UMP; all but the terminal —NH_2 and the attached carbon in carbamoylaspartate come from aspartate.

Biosynthesis and degradation of nucleotides
Page: 915 Difficulty: 3

64. Draw the structure of deoxythymidylic acid (dTMP). Indicate the source of each N and each C in the thymine ring, including its substituents.

    Ans: The structure of dTMP is shown in Fig. 22-47. The methyl substituent on the ring is derived from N^5,N^{10}-methylene tetrahydrofolate. The other atoms in the ring are derived from carbamoyl phosphate and aspartate, as seen in Fig. 22-38.

Biosynthesis and degradation of nucleotides
Pages: 915, 920Difficulty: 3

65. Describe the pathway by which GMP is converted into GTP; show the co-substrates that are involved and name the enzymes.

    Ans: GMP + ATP ⟶ GDP + ADP (nucleoside monophosphate kinase)
    GDP + ATP ⟶ GTP + ADP (nucleoside diphosphate kinase)

Biosynthesis and degradation of nucleotides
Page: 916 Difficulty: 1

66. Diagram the biosynthetic pathway from UMP to dTTP. Use abbreviations (e.g., UMP), not complete structures, and indicate where any co-substrates participate.

Ans:

UMP + ATP ⟶ UDP + ADP (nucleoside monophosphate kinase)
UDP + NADPH ⟶ dUDP + $NADP^+$ (ribonucleotide reductase and thioredoxin)
dUDP + ATP ⟶ dUTP + ADP (nucleoside diphosphate kinase)
dUTP ⟶ dUMP + PP_i (dUTPase)
dUMP + N^5,N^{10}-methylene tetrahydrofolate ⟶
dTMP + dihydrofolate (thymidylate synthase)
dTMP + ATP ⟶ dTDP + ADP (nucleoside monophosphate kinase)
dTDP + ATP ⟶ dTTP + ADP (nucleoside diphosphate kinase)

(See Fig. 22-46.)

Biosynthesis and degradation of nucleotides
Page: 916 Difficulty: 3

67. Show the reaction catalyzed by thymidylate synthase and explain with a simple diagram how the chemotherapeutic agents fluorouracil and methotrexate inhibit the synthesis of dTMP.

Ans: The reaction mechanism catalyzed by thymidylate synthase is shown in Fig. 22-53; dUMP is methylated on the pyrimidine ring, with N^5,N^{10}-methylenetetrahydrofolate as methyl donor, and dihydrofolate as the other product. The two key events of this mechanism are the nucleophilic attack at C6 of the pyrmidine ring and the removal of the proton on C5 of the pyrimidine ring. The modes of action of fluorouracil and methotrexate are shown in Fig. 22-52 and 22-53, p. 895. Fluorouracil is a suicide substrate of thymidylate synthase: The reaction begins as for the real substrate, but the enzyme gets stuck when trying to remove the –F instead of an $–H^+$. Methotrexate, on the other hand, is a potent competitive inhibitor of dihydrofolate reductase, thus preventing the regeneration of tetrahydrofolate.

Biosynthesis and degradation of nucleotides
Pages: 923–924 Difficulty: 3

68. Explain how lack of adenosine deaminase leads to a 100-fold increase in dATP, yet also causes a deficiency in dNTPs required for replication of T-cells and B-cells required for a functioning immune system.

Ans: Adenosine deaminase is a critical enzyme in the degradation of purines. A deficiency in this enzyme leads to extremely high concentrations of dATP. Instead of promoting DNA synthesis, these high concentrations of dATP actually shut down ribonucleotide reductase (see Fig. 22-44), thus preventing the formation of dCTP, dTTP, and dGTP. Therefore, there are no dNTPs available for DNA biosynthesis.

Biosynthesis and degradation of nucleotides
Page: 922 Difficulty: 3

69. Azaserine is a structural analog of glutamine. It is a competitive inhibitor of many enzymes that use glutamine as substrates. Name) three biosynthetic products whose synthesis you would expect to be inhibited by azaserine. Do you think that eating azaserine would be immediately fatal? Why or why not?

Ans: Azaserine (shown in Fig. 22-51) inhibits enzymes involved in the transfer of the amido nitrogen of glutamine in many biosynthetic pathways, including those leading to CTP, AMP, GMP, and tryptophan. Glutamine amidotransferases are in Fig. 22-10. Eating azaserine would not be immediately fatal because it would take some time for the deficiencies of CTP, AMP, GMP, tryptophan, etc., to become severe enough to cause death.

Biosynthesis and degradation of nucleotides
Page: 923 Difficulty: 3